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Palaeomagnetism of the Mesozoic Serra Geral Formation, southern Brazil

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ABSTRACT

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Palaeomagnetic results from 20 volcanic sequences and 11 intrusive bodies (sills and dykes) of the Serra Geral Formation (Paraná Basin) are reported in this paper. The sequences are widespread all over the basin, while sills and dykes (Ponta Grossa arch) come from the northeastern portion. Three mean palaeomagnetic poles were computed for the Serra Geral Formation, which account for a time interval of ~ 15 Ma. Pole SG1 is located at 85°S,108°E ($\alpha_{95} = 1.1°$, $N = 18$) and represents the main phase of the magmatic activity in the basin, with a mean age of ~ 135 Ma. Pole SG2 is located at 82°S,38°E ($\alpha_{95} = 7.8°$, $N = 2$) and represents a younger magmatic phase, with an associated age of ~ 130 Ma. Pole SG3 (72°S,37°E; $\alpha_{95} = 6.8°$, $N = 10$) is the youngest pole. It is computed from the intrusive rocks and its age is assumed to be not younger than 118 Ma, the lower limit of the 'Cretaceous normal magnetic interval'.

These three poles describe a shifting path, which suggests that the South American platform moved ~ 5° southwards and rotated ~ 10° clockwise during the Lower Cretaceous, preceding the South Atlantic opening.

1. Introduction

The Cretaceous poles from South America are not uniformly distributed, instead they delineate an elongated pattern that suggests a complex plate trajectory (e.g. Valencio et al., 1983) describing loops and 'hair pins' if individual poles are taken instead of mean poles. The reality of this swinging pole behaviour could be tested through a succession of reliable Cretaceous poles with good stratigraphic and age controls. In this way, an extensive palaeomagnetic analysis of the Serra Geral Formation (southern Brazil) is important since it corresponds probably to the most expressive continental magmatism in the world. This magmatic activity is believed to have lasted for many millions of years, and therefore, if the rapid polar shifts during the Cretaceous are true, the palaeomagnetic poles from Serra Geral formation should be able to detect them.

The palaeomagnetic survey of the Serra Geral Formation was first carried out by Creer (1962). Samples were collected over a large area and included extrusive and intrusive rocks. The scatter of site-mean directions of magnetization was attributed mainly to secular variation of the ancient geomagnetic field. A different approach in the palaeomagnetic investigations of these lavas was employed by Pacca and Hiodo (1976) and later by Ernesto et al. (1979), through a flow-wise sam-

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pling of four successions exposed on the steep southeastern fringes of the volcanic plateau. These results have been re-examined in the light of some additional experimental work and were presented by Bellieni et al. (1983). These investigations have since been extended to other regions throughout the basin, generally following the same field procedures. In this paper results will be presented from 20 lava-flow sequences and 11 intrusive bodies (sills and dykes) of the Serra Geral Formation, which enable a better understanding of the drift of South America during Lower Cretaceous time.

2. Geological setting

The Serra Geral Formation represents one of the largest known volumes of continental flood volcanics, covering about 1 200 000 km^2 of the

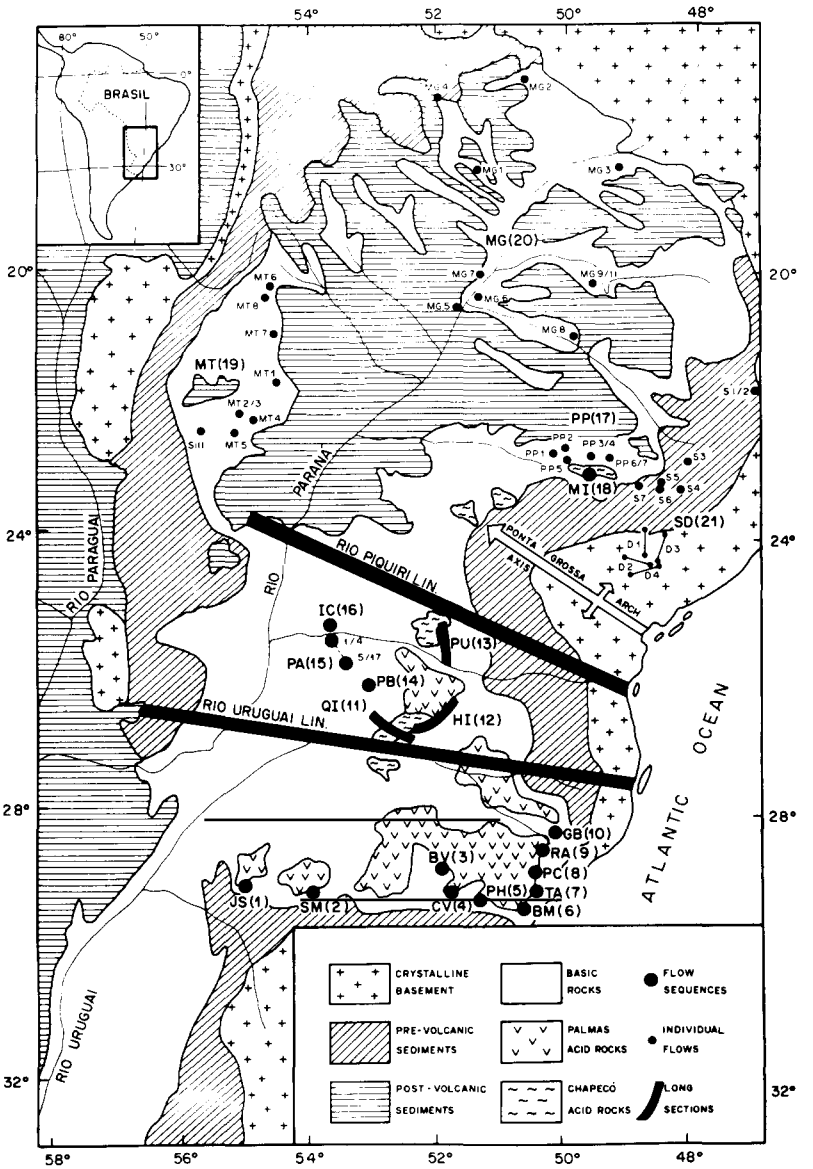


Fig. 1. Generalized geological map of the Paraná Basin showing the sampling sites.

Paraná Basin, mainly in southern Brazil but also in Uruguay, Paraguay and Argentina (Fig. 1). These lavas are usually over a kilometre thick; the maximum observed thickness being ~ 1700 m at the drill hole of Cuiaba Paulista, State of São Paulo (Almeida, 1986). Tholeiitic basalts are the dominant rock type, however, volcanic suites may also include acid flows, which occur mostly towards the eastern fringes of the basin. Sills and dykes are also found in the basin but generally concentrated to the northeastern parts, where the former preferentially intrude the Palaeozoic sediments.

The Serra Geral Formation lies over the Botucatu Formation (Jurassic–Cretaceous), which is composed of typical aeolian sandstones representing the top of the Gondwana sequence here. In the northern part of the basin the volcanics are overlain by Upper Cretaceous sediments, mainly the Bauru Formation. K–Ar and Rb–Sr ages indicate that the volcanism took place mainly during the Lower Cretaceous from 140 to 120 Ma (Creer et al., 1965; McDougall and Rüegg, 1966; Amaral et al., 1966; Melfi, 1967; Minioli et al., 1971; Sartori et al., 1975; Bossi and Umpierre-Urquhart, 1976; Siedner and Mitchell, 1976; Cordani et al., 1980; Mantovani et al., 1985; Piccirillo et al., 1987). The radiometric data show a rather large scatter, however; the K–Ar ages concentrate around a mean of 130 Ma with a standard deviation of ~ 15 Ma. The available data do not point clearly to any age province although Melfi (1967) suggests an older age (~ 130 Ma) for the western regions of the basin while younger ages (up to ~ 120 Ma, Amaral et al., 1966) preferentially apply to the volcanics of the eastern border.

Tectonic lineaments, e.g. the Rio Uruguai and Rio Piquiri lineaments in Fig. 1, cut the basin from the eastern edge to the centre. They are believed to have evolved since the Devonian and were particularly active during Triassic–Jurassic times (Fulfaro et al., 1982). Subparallel to them are the axes of arch-type structures. The most outstanding of them, the Ponta Grossa arch, is also characterized by pronounced NW–SE magnetic anomalies (Ferreira, 1982) and is intruded by hundreds of diabase dykes trending preferentially subparallel to the arch axis.

The recent geological and petrological data, summarized by Piccirillo et al. (1988a), suggest a subdivision of the basin according to the distribution and nature of the different volcanic rock types.

In the southern Paraná Basin (SPB; south of the Rio Uruguai lineament), basalts are characterized by low titanium content (< 2 wt.%). The volcanic suites in this region may include in their upper part, sequences of acid flows (Palmas type) comprising rhyodacites and rhyolites, sometimes with intercalations of basic lava flows. The classification adopted here is according to De la Roche et al. (1980). The acid sequences are mainly confined to the southeasternmost part of the basin and their thickness tends to increase towards the eastern border (Bellieni et al., 1986).

The northern Paraná Basin (NPB; north of the Rio Piquiri lineament) is dominated by tholeiites with high titanium content (> 2 wt.%) that are sometimes overlain by porphyritic rhyodacite flows (Chapecó acid type) towards the southeasternmost areas.

In the central Paraná Basin (CPB) the volcanic suites are similar to those occurring in the SPB and NPB, and may include both Palmas and Chapecó acid types. Here again acid flows tend to concentrate towards the eastern edge of the basin.

3. Sampling procedures

Samples for the palaeomagnetic work were collected essentially along road cuts. Sampling site locations can be seen on the simplified map of Fig. 1. Generally, three oriented hand samples were taken per outcrop, except for section BM(6) where a portable driller has been used. A total of nearly 1500 samples have been collected. Whenever possible, both magnetic and sun compasses were used for sample orientation.

The best sampling regions are in the southern border of the basin where the volcanic sequences are well exposed along the steep slopes. Within little more than 10 km, roads may cut through up to 1000 m of lava, starting at the pre-volcanic Botucatu sandstones until the top of the sequence. Ten sequences were sampled in this region, and

excepting the GB(10) section, which is composed exclusively of basalts, all of them include both basic and acid (Palmas type) rocks.

In the central Paraná region, topography is irregular due to vertical movements of faulted blocks and, as a consequence, different parts of the lava sequences are exposed along 100–200 km. When the stratigraphy was uncertain, the whole section has been subdivided into subsections corresponding to different blocks and referred to by roman numerals. The western sections PB(14), PA(15) and IC(16) exhibit essentially basalts, while QI(11), HI(12) and PU(13) include Chapeco acid volcanics. The HI section includes also Palmas acid rocks. A detailed description of these three sections, including chemical and palaeomagnetic data can be found in Nardy (1988).

Good exposures of lava sequences are somewhat scarce northwards, where the sampling sites are consequently spread over large areas. These flows were tentatively grouped into sequences based on the relative elevations (see Tables 17–20). Excepting section MI(18), a succession of four Chapeco acid flows, all the samples are from basic rocks.

Flows are generally flat-lying, but in the lower part of the sequences dips ranging from 8 to 25° can be found, which are probably due to the palaeotopography. The only section where tectonic tilting is really apparent is TA(7) where flows dip about 15° towards N30°W.

The studied sills are concentrated in the northeastern part of the basin and intrude Palaeozoic sediments. The dykes are subvertical and cut the Precambrian basement of the Ponta Grossa arch (Fig. 1).

4. Analytical work and results

At least three specimens (2.5 cm in diameter and 1.0 cm in height) from each sample were analysed. The remanent magnetizations were measured in a Digico spinner magnetometer. The samples were submitted to both a.f. and thermal demagnetization. The a.f. cleaning was performed

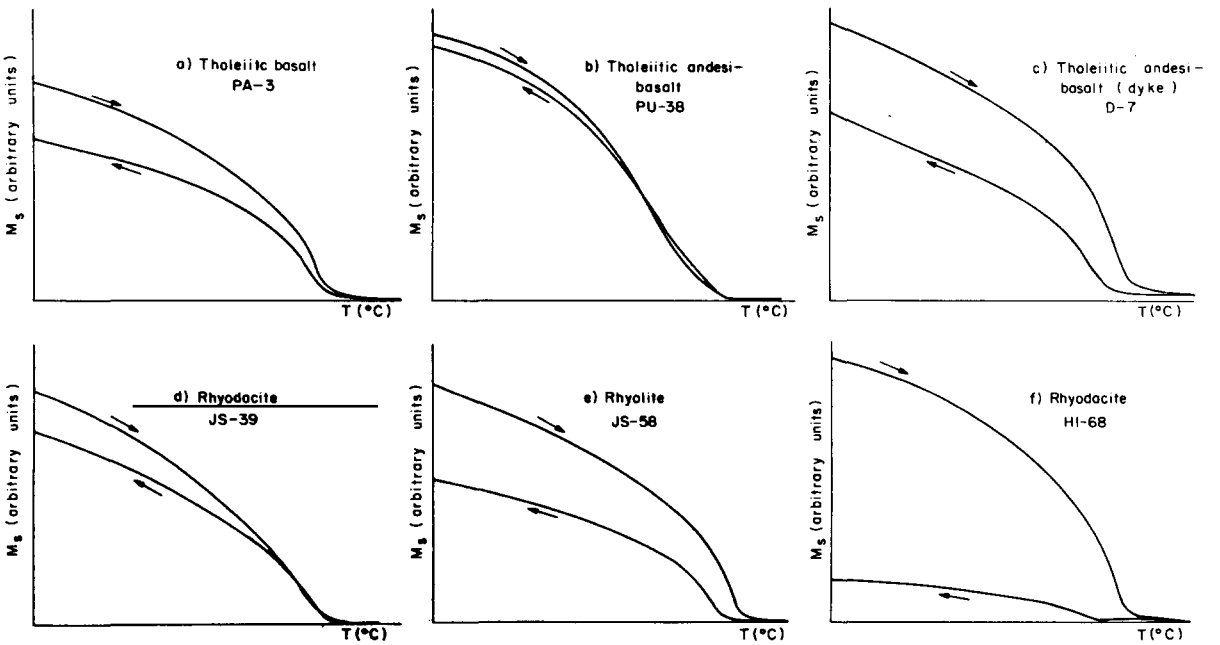


Fig. 2. Examples of the most common thermomagnetic curves obtained for the Serra Geral samples. The heating rate was 50°C min^{-1} in all cases. Classification of the lithologic types is according to De la Roche et al. (1980).

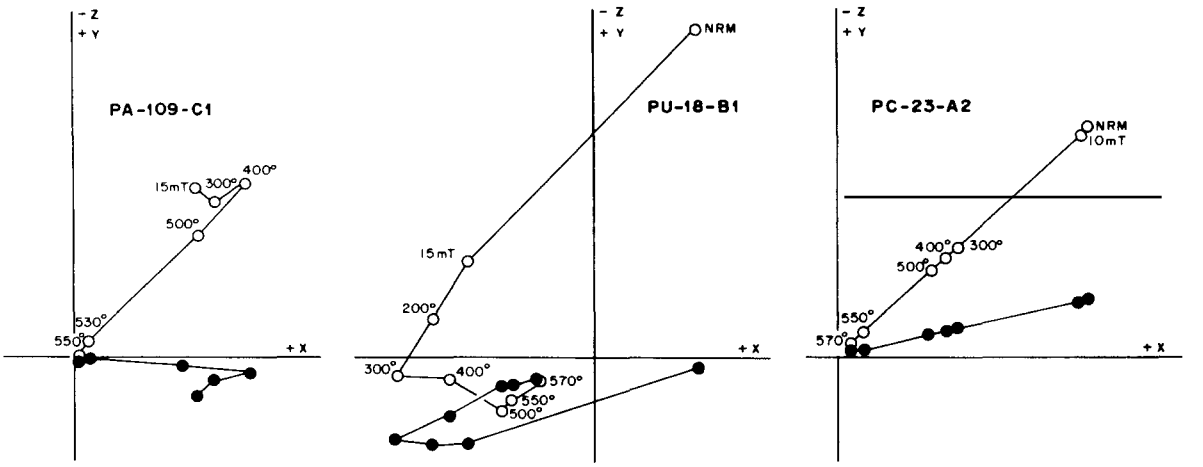


Fig. 3. Zijderveld plots for some of the studied samples.

from fields of 5 mT up to 80 mT in steps of 5 mT. Samples were thermally demagnetized at least at temperatures of 300, 400, 450, 500, 530, 550 and 570 °C; whenever necessary a more detailed procedure was adopted. An a.c. field of 10 mT or higher was applied before the thermal cleaning in order to remove the softer viscous remanent magnetizations. In general, the remanence was erased at temperatures slightly above 570 °C. Few samples retained part (< 10%) of their remanence at

TABLE 1

Palaeomagnetic data for sequence JS(1)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
20	440	3	188.7	27.6	2.8	1920.7	R	156.3	−73.3	1.7	3.1
19	430	3	187.7	31.6	8.9	188.7	R	157.4	−75.9	5.6	10.0
18	425	3	185.5	27.6	4.9	616.0	R	145.9	−74.5	2.9	5.3
17	420	6	196.7	29.7	4.7	200.2	R	178.3	−69.6	2.9	5.2
16	410	5	184.4	36.4	3.7	413.3	R	150.5	−80.1	2.5	4.3
15	395	3	192.7	35.4	6.2	395.1	R	178.5	−74.9	4.1	7.2
14	375	6	191.7	40.6	6.1	171.2	R	188.2	−77.9	3.7	6.2
13	365	5	191.8	32.2	7.1	114.2	R	170.8	−74.0	4.5	8.0
12	340	5	180.9	33.7	3.4	487.4	R	130.1	−79.2	2.2	3.9
11	330	3	168.2	46.8	12.4	98.8	R	39.0	−79.5	10.3	16.0
10	320	3	163.5	41.3	3.4	1306.1	R	52.0	−74.2	2.5	4.1
9	300	3	157.7	34.1	8.2	222.5	R	57.8	−67.1	5.4	9.4
8	290	3	175.6	36.4	10.8	131.1	R	100.5	−80.1	7.3	12.8
7	285	3	171.0	40.7	8.3	219.2	R	69.7	−79.9	6.1	10.1
6	255	3	346.0	−48.0	3.8	1029.7	N	212.8	77.7	3.2	5.0
5	220	9	344.9	−36.4	6.9	55.1	N	245.3	73.6	4.7	8.0
4	210	3	353.4	−49.5	8.5	209.1	N	202.5	84.1	7.5	11.3
3	190	9	349.8	−44.4	3.6	204.5	N	232.0	80.4	2.8	4.5
2	170	3	343.9	−49.3	7.9	242.5	N	207.5	75.9	6.9	10.5
1	150	4	359.8	−32.3	10.8	73.0	N	304.5	78.3	6.9	12.2
Sill	290	2	30.5	−51.0	32.4	61.2	N	48.7	63.6	29.6	43.8

TABLE 2

Palaeomagnetic data for sequence SM(2)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
20	500	3	166.2	41.6	11.2	121.5	R	57.1	−76.5	8.4	13.7
19	480	3	168.8	47.3	13.9	78.6	R	39.0	−80.1	11.7	18.0
18	460	3	177.9	41.3	7.6	261.7	R	107.8	−84.0	5.6	9.3
17	415	8	179.5	35.8	4.8	132.2	R	123.9	−80.4	3.2	5.6
16	410	3	169.9	44.4	6.7	334.2	R	54.4	−80.5	5.3	8.4
15	400	4	184.5	31.8	11.1	69.1	R	146.4	−77.1	7.0	12.5
14	395	5	170.1	43.7	1.6	2202.5	R	57.7	−80.4	1.2	2.0
13	390	3	167.8	42.4	13.2	87.8	R	57.7	−78.1	10.0	16.3
12	385	3	164.4	33.0	11.0	126.3	R	71.8	−71.7	7.1	12.5
11	380	3	168.0	31.5	3.0	1640.5	R	82.4	−73.5	1.9	3.4
10	375	3	16.7	−32.2	20.2	38.2	N	2.4	70.6	12.8	22.8
9	360	3	359.8	−41.5	5.2	549.8	N	304.6	84.4	3.9	6.3
8	320	6	4.7	−42.8	6.4	108.7	N	350.2	83.8	4.9	7.9
7	315	3	356.9	−30.5	9.9	154.4	N	293.7	76.7	6.1	11.0
6	290	2	2.0	−36.0	37.8	45.7	N	318.4	80.3	25.5	43.9
5	280	4	6.8	−35.4	4.9	349.5	N	340.2	78.3	3.3	5.7
4	275	3	4.6	−37.3	11.8	109.4	N	233.5	80.4	8.1	13.9
3	270	2	163.3	52.8	31.4	65.5	R	16.7	−75.2	30.0	43.4
2	265	3	193.0	42.9	12.0	106.0	R	198.3	−77.6	9.2	14.9
1	260	3	357.5	−30.5	7.5	270.1	N	297.3	76.8	4.6	8.3
Sill	150	3	218.7	0.1	15.6	63.2	I	185.1	42.8	7.8	15.6

TABLE 3

Palaeomagnetic data for sequence BV(3)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
21	565	7	358.9	−44.0	3.8	251.9	N	292.0	86.7	3.0	4.8
20	560	3	355.4	−41.5	9.7	162.2	N	268.7	83.5	7.2	11.8
19	545	2	354.1	−48.7	35.7	50.8	N	208.5	84.8	31.0	47.0
18	530	2	359.1	−43.5	21.0	142.7	N	295.4	86.4	16.3	26.2
17	520	2	344.1	−31.7	38.5	44.0	N	254.2	71.4	24.3	43.2
16	510	3	355.0	−40.9	9.0	186.5	N	268.4	82.9	6.6	10.9
15	500	2	341.4	−47.4	11.9	436.2	N	215.1	73.7	10.0	15.4
14	485	3	2.5	−51.6	12.4	99.6	N	96.0	86.0	11.5	16.9
13	435	3	358.2	−41.3	11.1	122.7	N	291.6	84.5	8.2	13.5
12	405	3	4.7	−34.4	22.7	30.4	N	332.7	79.0	14.9	26.0
11	395	4	5.9	−42.8	9.8	88.1	N	2.1	83.3	7.5	12.1
10	370	3	1.7	−41.4	4.3	817.1	N	325.8	84.6	3.2	5.2
9	345	2	305.4	15.1	15.3	266.8	I	244.5	25.9	8.1	15.7
8	310	3	347.6	−45.5	7.3	281.8	N	225.6	78.9	5.9	9.3
7	275	3	174.9	18.5	8.0	238.1	R	113.6	−70.0	4.3	8.3
6	260	2	174.5	22.7	1.8	18565.2	R	110.6	−72.1	1.0	1.9
5	235	3	176.5	35.4	18.3	46.0	R	109.3	−79.9	12.2	21.2
4	220	2	352.3	−55.8	19.1	172.0	N	167.6	80.1	19.6	27.4
3	200	3	2.0	−33.8	12.9	91.7	N	319.2	79.4	8.4	14.7
2	170	3	4.8	−34.7	11.3	118.0	N	333.9	79.2	7.5	13.0
1	150	3	10.6	−40.0	11.4	116.7	N	8.3	78.6	8.3	13.7

TABLE 4

Palaeomagnetic data for sequence CV(4)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (° E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
24	660	3	345.2	−20.6	13.6	82.0	N	269.2	66.7	7.5	14.3
23	625	2	348.9	29.7	12.2	418.5	I	293.7	43.4	7.5	13.5
22	565	2	332.6	42.9	15.7	252.0	I	279.8	29.6	12.0	19.4
21	545	6	3.4	11.0	9.1	55.1	I	314.6	54.8	4.7	9.2
20	510	3	357.9	−41.5	6.5	354.3	N	289.5	84.1	4.8	7.9
19	470	3	352.5	−44.3	12.5	97.4	N	243.8	82.5	9.9	15.7
18	450	3	352.6	−23.9	3.1	1528.8	N	284.9	71.7	1.8	3.3
17	440	3	356.7	−23.3	6.8	327.6	N	297.8	72.4	3.9	7.2
16	425	3	195.0	35.0	9.9	153.7	R	185.5	−72.9	6.6	11.4
15	405	2	355.4	−40.4	5.9	1759.3	N	274.9	82.4	4.3	7.1
14	380	3	182.3	47.7	4.5	737.1	R	201.3	−87.8	3.8	5.9
13	360	3	193.8	56.9	7.4	277.7	R	257.4	−75.9	7.8	10.8
12	340	3	163.9	58.5	14.5	73.1	R	358.0	−73.5	15.9	21.5
11	320	3	348.7	−59.3	11.4	116.9	N	166.0	75.8	12.1	17.1
10	270	3	356.8	−37.7	10.4	140.0	N	288.9	81.2	7.2	12.3
9	230	3	345.7	−47.2	5.9	423.6	N	219.8	77.5	4.9	7.6
8	160	7	356.3	−44.3	4.5	179.4	N	263.8	85.3	3.5	5.7
7	150	2	352.3	−46.0	6.3	1529.2	N	233.1	82.9	5.1	8.1
6	135	2	351.1	−48.8	17.5	205.1	N	214.0	82.2	15.2	23.5
5	115	3	355.4	−47.2	10.2	146.3	N	232.4	85.9	8.6	13.2
4	90	3	357.3	−49.1	4.8	652.7	N	204.0	87.6	4.2	6.3
3	80	6	11.6	−34.9	4.4	227.1	N	357.3	75.2	2.9	5.1
2	70	3	10.9	−29.0	7.8	249.1	N	346.9	72.8	4.7	8.6
1	65	3	12.5	−41.0	8.1	227.5	N	13.7	77.3	6.0	9.8

TABLE 5

Palaeomagnetic data for sequence PH(5)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (° E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
18	600	4	178.4	29.5	9.4	94.8	R	122.6	−76.4	5.7	10.4
17	570	5	175.6	27.9	8.9	74.9	R	112.2	−74.9	5.3	9.7
16	560	3	178.6	39.5	5.5	501.2	R	118.3	−82.9	3.9	6.6
15	550	2	169.9	22.1	25.2	99.9	R	98.9	−69.8	14.1	26.7
14	520	4	180.2	35.9	6.7	185.7	R	130.1	−80.5	4.5	7.8
13	510	3	25.3	−78.6	16.4	56.9	I	114.9	48.4	29.4	31.0
12	440	3	192.2	62.5	6.6	348.0	R	278.3	−72.4	8.1	10.3
11	410	2	52.2	27.7	43.8	34.5	I	5.0	23.0	26.1	47.8
10	400	3	52.7	4.8	10.2	146.2	I	16.1	30.4	5.1	10.2
9	360	2	65.5	−32.2	23.3	116.9	I	43.3	29.4	14.8	26.3
8	340	2	344.1	−41.1	26.1	93.3	N	237.2	74.6	19.3	31.7
7	335	3	350.6	−51.3	14.2	76.3	N	198.4	81.4	13.1	19.3
6	310	2	346.4	−23.7	19.9	158.6	N	269.3	68.8	11.3	21.2
5	285	2	1.2	11.6	4.3	3367.9	I	311.0	54.7	2.2	4.4
4	265	3	179.4	32.5	11.3	119.6	R	126.3	−78.3	7.2	12.8
3	240	3	200.7	32.1	6.8	329.8	R	191.4	−67.6	4.3	7.7
2	110	3	20.2	−58.2	2.2	2984.8	N	74.1	70.7	2.4	3.2
1	90	3	10.8	−40.9	8.4	212.5	N	10.2	78.6	6.2	10.2

TABLE 6
Palaeomagnetic data for sequence BM(6)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
19	775	2	1.8	−41.0	28.5	79.1	N	325.7	85.0	21.0	34.6
18	730	3	356.0	−43.5	5.3	295.7	N	268.2	84.6	4.1	6.6
17	715	4	337.4	24.1	10.8	72.5	I	279.1	42.5	6.2	11.5
16	685	3	344.9	−19.7	19.5	41.2	N	270.3	66.2	10.7	20.4
15	675	3	333.0	−32.8	12.2	101.7	N	238.7	62.8	7.8	13.8
14	645	2	196.8	50.7	14.1	311.5	R	231.5	−75.3	12.8	19.0
13	565	2	184.0	36.7	32.9	59.5	R	152.7	−80.2	22.5	38.5
12	495	3	177.1	44.2	9.5	167.3	R	93.8	−85.6	7.5	11.9
11	450	4	128.4	49.0	12.9	51.6	R	25.6	−45.5	11.3	17.0
10	415	6	4.3	−56.4	7.0	91.0	N	104.9	81.6	7.3	10.1
9	380	5	352.4	−54.6	7.8	96.7	N	175.6	81.4	7.8	11.0
8	335	7	350.6	−47.0	6.9	75.4	N	226.3	81.7	5.8	8.9
7	255	6	354.2	−26.9	6.6	101.4	N	289.2	73.8	3.9	7.2
6	235	4	358.7	−31.5	9.3	97.4	N	304.1	77.5	5.8	10.4
5	225	3	163.1	21.1	10.9	128.5	R	85.7	−65.6	6.0	11.5
4	220	7	92.3	−62.9	4.1	209.9	I	80.7	18.6	5.1	6.4
3	210	4	165.3	19.5	10.6	75.9	R	91.3	−66.1	5.8	11.1
2	200	4	178.5	32.1	10.5	77.3	R	122.8	−77.8	6.7	11.8
1	190	3	168.2	7.0	14.0	78.1	R	104.0	−61.7	7.1	14.1

TABLE 7
Palaeomagnetic data for sequence TA(7)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
19	870	2	1.3	−41.8	52.3	24.9	N	322.6	84.5	39.2	64.1
18	835	3	344.0	−32.4	21.4	34.0	N	255.2	71.3	13.6	24.2
17	785	3	353.8	−31.1	16.6	55.9	N	284.3	76.2	10.4	18.6
16	735	3	177.4	12.6	7.7	251.8	R	123.3	−66.8	4.0	7.8
15	715	3	335.7	−54.2	13.2	88.0	N	198.9	68.7	13.0	18.6
14	630	5	337.2	−64.0	4.2	330.1	N	170.9	65.7	5.3	6.7
13	535	3	350.5	−42.7	10.5	137.2	N	246.0	80.4	8.0	13.0
12	510	3	358.7	−34.5	15.8	61.3	N	303.1	79.5	10.4	18.1
11	490	3	17.2	−34.0	20.3	37.7	N	9.3	70.9	13.3	23.2
10	450	3	164.9	38.2	14.3	75.1	R	66.5	−74.3	10.0	16.9
9	415	3	183.5	63.5	5.1	571.1	R	300.7	−74.0	6.4	8.1
8	405	3	52.9	−39.5	13.4	85.2	I	44.3	42.2	9.6	16.1
7	370	4	353.9	−37.7	10.9	70.9	N	275.0	80.1	7.6	12.9
6	290	2	356.1	−15.5	10.9	525.1	N	299.4	68.2	5.8	11.2
5	280	5	354.9	−49.6	4.9	237.6	N	204.8	85.5	5.5	5.8
4	255	3	356.7	−36.4	5.5	499.3	N	290.9	80.4	3.7	6.4
3	230	3	349.6	−36.5	6.3	383.4	N	261.6	76.9	4.3	7.4
2	175	6	280.9	−52.0	5.1	168.8	I	194.5	23.8	4.8	7.0
1	120	3	237.0	−44.9	9.8	158.7	I	179.9	−11.7	7.8	12.4

TABLE 8

Palaeomagnetic data for sequence PC(8)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
27	900	4	7.5	−43.1	17.5	28.2	N	11.1	82.1	13.5	21.7
26	883	2	277.7	−44.4	49.8	27.2	I	199.8	18.7	39.3	62.6
25	880	2	260.4	−40.4	50.3	26.7	I	195.2	3.2	36.7	60.8
24	845	4	3.7	−45.9	5.3	298.7	N	12.0	86.2	4.3	6.8
23	825	4	358.2	−43.2	8.7	110.3	N	288.1	85.8	6.7	10.8
22	810	2	4.0	−39.4	5.4	2140.4	N	339.1	82.3	3.9	6.5
21	800	3	0.4	−43.0	12.6	96.0	N	315.6	85.8	9.7	15.6
20	775	3	359.6	−46.1	8.0	233.3	N	299.9	88.3	6.6	10.2
19	735	3	351.9	−44.5	9.5	167.9	N	240.3	82.2	7.5	12.0
18	725	2	353.0	−45.7	16.6	227.3	N	235.8	83.5	13.5	21.2
17	700	3	184.1	59.8	31.3	16.5	R	294.8	−77.9	35.6	47.2
16	650	5	356.4	−35.4	9.9	60.4	N	290.4	79.9	6.6	11.4
15	580	2	351.1	−38.7	20.9	144.7	N	260.3	79.1	14.8	24.9
14	570	3	1.1	−19.3	11.6	113.9	N	313.3	70.8	6.3	12.1
13	560	2	353.5	−45.5	3.3	5633.2	N	238.6	83.8	2.7	4.2
12	535	2	332.4	−33.2	5.1	2354.3	N	237.4	62.5	3.3	5.8
11	500	3	3.5	−41.1	7.7	253.8	N	340.7	83.6	5.7	9.4
10	435	3	355.2	−32.7	15.0	67.7	N	287.7	77.9	9.6	17.0
9	405	4	356.2	−40.0	5.8	248.2	N	281.1	82.8	4.2	7.0
8	350	2	174.5	1.2	34.6	54.0	I	118.6	−61.0	17.3	34.6
7	270	3	353.6	−34.0	13.4	84.9	N	279.3	78.0	8.8	15.3
6	230	2	7.4	−43.5	30.5	68.7	N	12.4	82.3	23.7	38.0
5	220	3	341.5	−48.8	8.1	228.2	N	213.1	73.9	7.0	10.7
4	170	4	2.8	−53.4	13.2	48.7	N	104.1	84.5	12.8	18.4
3	140	2	175.6	59.6	26.6	90.2	R	326.1	−78.0	30.1	40.0
2	115	4	349.5	−50.3	4.5	402.8	N	205.0	80.7	4.0	6.3
1	70	3	340.8	−45.1	6.0	419.2	N	223.4	72.9	4.8	7.6

temperatures up to 670 °C. Possible changes in susceptibility values were monitored using a Bison susceptibility meter.

Typical thermomagnetic curves (Fig. 2) for samples from different sites and lithologies show high Curie temperatures ranging from 540 to 595 °C, indicating that the main magnetic carriers are highly oxidized titanomagnetites. Ore microscope observations also show that titanomagnetites in most of the rocks underwent high-temperature oxidation. Although any of the six types of thermomagnetic curves shown in Fig. 2 could be found indistinctly in each of the lithologic groups, type f were more frequent in acid rocks. This curve seems to be related to a higher content of maghemite, possibly due to weathering. Generally, samples of this type gave no consistent results and were discarded.

The demagnetization data have been analysed by means of Wulff stereograms and Zijderveld plots (Zijderveld, 1967), as seen in Fig. 3. The majority of samples showed a single component of magnetization after the elimination of a soft viscous remanence by applying an a.c. field not greater than 15 mT. The other samples displayed multicomponent magnetizations, which could not always be resolved. About 30% of the total number of samples were rejected owing to unreliable results. The mean magnetization direction of each hand sample or core was obtained by the average of at least three specimens (two specimens for a very few samples). The site (one flow or intrusive

TABLE 9

Palaeomagnetic data for sequence RA(9)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
19	1060	2	8.2	−54.5	21.1	141.8	N	3.9	81.7	21.0	29.8
18	1025	2	5.4	−29.3	36.0	50.1	N	332.0	75.9	21.9	39.7
17	995	3	348.0	−39.8	6.4	369.2	N	247.1	77.5	4.6	7.7
16	940	2	346.2	−36.6	23.5	114.5	N	250.9	74.9	16.0	27.4
15	915	3	4.1	−38.7	18.9	43.5	N	339.1	82.1	13.4	22.5
14	900	3	2.4	−38.1	15.4	64.7	N	327.1	82.2	10.8	18.2
13	845	2	5.9	−54.6	29.2	75.1	N	93.2	81.9	29.1	41.2
12	820	6	351.7	−44.4	8.3	65.7	N	238.3	82.1	6.6	10.4
11	805	3	347.3	−64.0	5.0	585.9	N	157.2	70.3	6.3	8.0
10	780	7	353.3	−49.8	5.0	145.0	N	274.2	80.1	4.4	6.7
9	690	3	1.2	−57.2	17.2	52.2	N	124.0	80.9	18.3	25.1
8	655	4	353.6	−50.2	2.2	1664.1	N	197.1	84.0	2.0	2.9
7	620	3	209.2	56.3	17.4	51.0	R	245.9	−64.2	18.1	25.1
6	575	7	353.9	−31.9	7.4	66.0	N	282.8	77.1	4.7	8.3
5	540	3	224.7	−9.5	5.0	589.1	I	189.4	−35.4	2.6	5.1
4	495	2	59.3	−51.9	56.5	21.6	I	60.0	39.5	52.7	77.2
3	420	3	12.8	−34.3	13.7	80.8	N	2.4	74.6	9.0	15.7
2	370	3	358.4	−38.5	9.6	165.3	N	298.2	82.7	6.8	11.4
1	330	3	1.2	−26.3	3.7	1095.5	N	314.7	75.0	2.2	4.0

body) mean directions were calculated giving unit weight to each sample.

Tables 1–21 display the palaeomagnetic results for all volcanic sets (flow sequences or intrusive bodies). The identified flows are numbered in a sequential order from bottom to top. Flow identification, whenever possible, was based on geological, geochemical (Piccirillo et al., 1988a) and palaeomagnetic data. In these tables, N is the number of samples used in the means, α_{95} and K are Fisher's (1953) statistical parameters; d_p and d_m define the 95% confidence circle for the virtual geomagnetic poles (VGPs). Both normal and reversed polarities are present in almost all the sequences and intrusive bodies.

The mean magnetization direction and the corresponding palaeomagnetic pole for each volcanic set were calculated giving unit weight to each site. However, sites with $\alpha_{95} > 30°$ were rejected for calculations, as well as supposedly transitional directions associated with a reversing field (Table 22). The cut off α_{95} value of 30° was adopted based on the results reported by Ernesto and Pacca (1988). According to these authors, within a thick flow (sometimes actually multiflows) the magnetization may vary considerably, depending on the cooling time, given rise to high scattering of the remanence directions.

5. Magnetic signature in the Serra Geral flow sequences

Changes in magnetic declination and inclination for each sequence from the southern region are shown in Figs. 4 and 5, respectively (see pp. 170 and 171). Magnetic data are plotted against elevation, which allows account to be taken of possible missing flows in each sequence, which could not be sampled or did not give good results. These are the thickest studied sequences, all of which rest on the pre-volcanic sediments, and most of which comprise several polarity intervals. These intervals, however, are difficult to correlate because the volcanic activity may not have started simultaneously in different sequences, as seen in the records of Figs. 4 and 5. Furthermore, in a situation like that of the Lower Cretaceous, with a

TABLE 10

Palaeomagnetic data for sequence GB(10)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
29	1325	3	1.0	−42.3	7.4	274.0	N	323.5	85.9	5.6	9.1
28	1315	2	348.7	−38.1	9.0	768.1	N	252.4	77.6	6.3	10.6
27	1305	3	12.0	−36.3	7.3	279.6	N	6.2	76.3	5.0	8.5
26	1295	4	354.7	−49.1	5.3	296.5	N	200.3	85.1	4.6	7.0
25	1290	2	357.2	−40.7	9.3	720.9	N	283.8	84.2	6.8	11.3
24	1275	4	355.1	−41.2	8.6	114.0	N	266.7	83.5	6.4	10.5
23	1265	3	357.0	−33.9	11.6	113.2	N	294.3	79.7	7.6	13.2
22	1250	2	2.6	−53.9	8.5	858.3	N	111.0	83.5	8.3	11.9
21	1225	2	326.4	−46.8	3.1	6446.7	N	213.1	60.4	2.6	4.0
20	1210	2	312.0	−66.3	9.1	744.9	N	177.9	48.2	12.3	14.9
19	1200	3	354.1	−52.8	9.9	154.1	N	174.6	82.9	9.4	13.7
18	1195	2	349.1	−53.1	32.1	62.5	N	184.4	79.2	30.8	44.5
17	1185	3	14.8	−30.8	5.7	457.1	N	2.8	71.9	3.5	6.4
16	1175	2	4.2	−34.7	16.4	231.8	N	334.0	79.9	10.8	18.9
15	1150	3	8.1	−39.8	9.9	153.9	N	4.0	80.6	7.1	11.9
14	1140	3	0.5	−27.5	12.1	103.4	N	312.7	76.1	7.2	13.2
13	1135	3	4.3	−31.2	9.3	175.6	N	330.5	77.7	5.8	10.4
12	1110	2	354.8	−33.9	30.8	67.6	N	283.6	79.0	20.1	35.2
11	1085	2	358.0	−29.4	5.9	1776.3	N	301.8	77.2	3.6	6.5
10	1025	2	189.6	23.7	19.7	161.5	R	161.7	−71.6	11.2	21.0
9	1005	2	170.7	45.4	28.1	80.8	R	48.8	−81.6	22.7	35.7
8	990	2	176.3	47.7	15.5	261.5	R	32.8	−86.7	13.1	20.2
7	965	3	175.5	43.9	6.3	383.8	R	73.5	−85.1	4.9	7.9
6	955	2	179.1	48.0	8.3	905.2	R	1.1	−88.9	7.1	10.8
5	930	2	182.6	43.9	16.4	233.1	R	171.7	−86.4	12.8	20.5
4	911	3	188.7	48.6	6.8	326.7	R	231.3	−82.3	5.9	8.9
3	886	4	195.8	42.8	8.6	113.8	R	210.1	−75.4	6.6	10.6
2	841	2	178.4	29.4	26.6	90.2	R	123.6	−77.2	16.2	29.4
1	811	3	185.8	33.9	5.0	608.2	R	160.0	−78.8	3.3	5.7

high reversal frequency (Kent and Gradstein, 1985), it becomes difficult to correlate polarity intervals unless a stratigraphic marker exists.

The southern sequences are thus tentatively put in a chronological order by comparing the declination and inclination records in Figs. 4 and 5, respectively. Sequences seem to be best correlated when the short normal polarity event in sequences (3)–(6) is used as a stratigraphic marker (dashed lines in the figures) as well as the prominent declination change in sequences PC(8) and GB(10). Detailed correlations of component flow directions in different lava sections were shown by Ernesto (1985), suggesting that these matches are not chance correlations. Furthermore, the mean magnetization directions, after eliminating the effects of geographical distances, are statistically similar when compared using an F test, as proposed by McFadden and Lowes (1981), at a 95% confidence level. Nevertheless this is not true for the westernmost sequences JS(1) and SM(2); their mean magnetizations are similar in respect of one to the other, but discordant from any other in the basin, at the same confidence level.

It is interesting to note on the plots of Figs. 4 and 5 that although parts of two or more sequences seem to be formed at the same time, flows are not likely to be the same even in sequences that are as close as 30 km, for instance of sequences CV(4) and PH(5), because lithologies are frequently different.

Figures 6 and 7 (p. 172) show the same sort of

TABLE 11

Palaeomagnetic data for sequence QI(11)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
I-3 [a]	1040	4	347.4	−50.4	4.8	363.8	N	194.3	78.2	4.3	6.4
I-2 [a]	1015	3	346.9	−40.6	6.1	409.0	N	232.4	77.5	4.5	7.4
I-1 [a]	1010	2	356.5	−47.4	16.2	237.2	N	189.4	86.4	13.6	21.0
II-1	1015	3	344.9	−36.7	16.3	57.8	N	239.8	74.7	11.1	19.0
III-2	1060	3	340.7	−55.8	9.9	155.6	N	183.2	71.0	10.2	14.2
III-1 [a]	970	3	345.3	−41.2	11.0	124.6	N	228.4	76.3	8.2	13.4
IV-1 [a]	1005	2	356.7	−54.6	19.0	174.2	N	146.0	81.2	18.9	26.8
V-3	985	3	7.6	−41.0	16.1	59.6	N	13.4	82.3	11.9	19.6
V-2 [a]	805	4	341.6	−48.0	5.8	248.9	N	205.7	73.6	5.0	7.6
V-1 [a]	715	4	341.7	−44.5	9.4	96.2	N	215.9	73.6	7.4	11.8
VI-3	820	6	354.2	−52.8	4.4	229.3	N	164.0	81.8	4.2	6.1
VI-2	810	3	352.3	−39.2	17.9	48.4	N	249.8	81.5	12.8	21.4
VI-1 [a]	785	2	350.2	−55.2	33.6	57.1	N	168.6	77.8	34.0	47.8
VII-9	790	5	354.8	−47.1	5.8	169.6	N	199.6	85.1	4.9	7.5
VII-8	770	3	4.8	−43.3	11.3	119.1	N	16.9	85.3	8.7	14.0
VII-7	755	6	2.2	−44.7	5.5	145.2	N	21.0	87.9	4.4	6.9
VII-6	715	6	359.0	−40.7	2.6	658.9	N	293.1	86.2	1.9	3.2
VII-5	655	6	357.9	−45.4	5.9	129.7	N	217.8	88.1	4.8	7.5
VII-4	575	3	197.8	35.3	7.6	261.3	R	196.6	−72.0	5.1	8.8
VII-3	470	6	168.7	35.8	3.9	294.1	R	69.3	−77.4	2.6	4.5
VII-2	430	4	176.9	41.0	19.7	22.6	R	87.3	−85.5	14.5	23.9
VII-1	365	9	359.2	−47.4	5.6	84.1	N	150.8	88.2	4.7	7.3

[a] Chapecó acid-type flows.

TABLE 12

Palaeomagnetic data for sequence HI(12)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
I-6	1305	4	185.9	45.8	8.4	119.6	R	226.3	−84.7	6.8	10.7
I-5	1275	3	189.3	36.4	13.8	80.3	R	183.6	−79.3	9.4	16.1
I-4	1265	3	195.0	37.6	15.5	63.9	R	199.5	−75.2	10.8	18.3
I-3	1245	3	190.0	30.2	10.2	146.8	R	172.2	−76.0	6.3	11.3
I-2	1235.	3	171.7	33.1	18.3	46.1	R	84.8	−78.5	11.8	20.8
I-1	1190	1	189.5	4.6	–	–	R	150.5	−64.0	–	–
II-1	1235	4	179.1	26.7	14.4	41.4	R	124.2	−77.3	8.5	15.6
III-2	1260	3	178.5	33.9	32.8	15.1	R	118.3	−81.6	21.4	37.5
III-1	1210	6	185.0	34.0	4.1	263.1	R	158.6	−80.6	2.7	4.7
IV-4	1190	3	190.4	26.9	10.5	138.5	R	167.8	−74.0	6.2	11.4
IV-3	1100	3	165.9	50.2	25.6	24.1	R	16.3	−76.9	23.0	34.3
IV-2	1070	3	164.8	51.5	15.0	68.3	R	12.6	−75.7	13.9	20.4
IV-1	1040	3	161.0	43.6	7.6	263.2	R	38.4	−72.9	5.9	9.5
V-5	1120	3	165.4	13.6	9.1	184.4	R	90.6	−65.7	4.7	9.3
V-4	1100	3	177.6	26.3	8.6	202.9	R	117.8	−76.8	5.0	9.3
V-3	1080	3	172.3	45.8	25.3	24.7	R	33.2	−83.1	20.6	32.3
V-2	1050	4	173.6	43.9	15.8	24.1	R	47.9	−84.1	12.4	19.8
V-1	1035	3	174.7	38.8	7.2	291.4	R	82.6	−83.1	5.1	8.6
VI-5 [a]	1125	6	353.1	−45.9	10.5	53.6	N	212.6	83.8	8.6	13.4
VI-4 [a]	1080	3	340.9	−40.1	25.3	24.6	N	227.0	72.2	14.3	26.5
VI-3	1065	3	160.0	42.0	4.8	649.5	R	41.9	−71.7	3.6	5.9
VI-2	1060	2	181.5	30.8	10.3	581.4	R	136.0	−79.6	6.4	11.5
VI-1	1035	3	190.4	46.1	9.9	155.0	R	224.0	−80.7	8.1	12.7

[a] Chapecó acid-type flows.

TABLE 13

Palaeomagnetic data for sequence PU(13)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
I-7 [a]	1035	3	344.6	−40.4	3.4	1254.7	N	224.9	75.7	2.5	4.1
I-6 [a]	1020	2	332.8	−53.6	28.7	77.6	N	191.7	64.9	27.9	40.0
I-5 [a]	1005	5	343.3	−35.6	12.3	39.3	N	235.5	73.5	8.2	14.2
I-4	975	3	161.5	20.3	14.0	78.2	R	75.6	−66.9	7.7	14.7
I-3	970	7	169.9	24.6	6.3	90.2	R	89.4	−74.2	3.6	6.8
I-2	950	5	167.4	25.4	11.6	44.3	R	81.7	−73.0	6.7	12.5
I-1	925	5	162.8	25.1	9.0	72.1	R	72.4	−69.6	5.2	9.7
II-5 [a]	1020	6	348.8	−48.4	3.3	398.2	N	195.4	79.4	2.8	4.3
II-4 [a]	955	4	356.1	−44.6	16.1	33.4	N	206.8	86.4	12.8	20.3
II-3	915	2	171.8	26.4	14.4	299.6	R	93.4	−76.0	8.5	15.6
II-2	865	6	162.0	24.6	4.8	194.6	R	71.9	−68.8	2.8	5.1
II-1	855	3	159.2	34.0	22.2	31.8	R	54.1	−69.5	14.5	25.4
III-2 [a]	1170	3	350.9	−34.8	10.6	134.2	N	254.8	79.2	7.0	12.2
III-1 [a]	1120	3	358.0	−55.9	6.3	373.8	N	37.1	79.2	6.5	9.0
IV-11 [a]	1130	3	358.8	−58.0	7.8	250.3	N	132.8	77.0	8.5	11.5
IV-10	1100	2	180.6	19.9	62.7	18.0	R	130.8	−74.4	34.3	65.6
IV-9	1040	3	165.3	20.9	5.1	565.5	R	83.0	−69.5	2.8	5.4
IV-8	1020	3	170.4	23.9	18.7	44.4	R	92.5	−73.9	10.7	20.0
IV-7	970	3	356.9	−44.0	15.9	61.1	N	217.4	87.2	12.5	19.9
IV-6	910	7	0.2	−48.9	3.8	249.3	N	126.1	85.9	3.3	5.0
IV-5	870	3	357.3	−40.0	15.6	63.1	N	268.6	86.1	11.3	18.8
IV-4	830	3	1.1	−45.3	6.7	331.8	N	85.8	88.5	5.4	8.5
IV-3	750	3	18.7	−32.7	27.6	20.9	N	17.7	70.9	17.7	31.2
IV-2	725	3	356.9	−30.4	19.4	41.3	N	290.9	80.1	12.0	21.6
IV-1	650	2	14.2	−40.9	52.3	24.8	N	31.4	76.4	38.5	63.5

[a] Chapecó acid-type flows.

plots for sequences from central and northern regions. These sequences, however, are not so thick and display three polarity horizons at most. The reversed flows are therefore tentatively correlated. The most striking feature in these plots is that the Chapecó-type acid flows that occur on the top of sequences QI(11), HI(12) and PU(13), which are subdivided into blocks, are all of normal polarity.

TABLE 14

Palaeomagnetic data for sequence PB(14)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
9	880	3	342.2	−43.4	8.8	193.8	N	216.2	73.9	6.8	10.9
8	850	2	315.4	−48.7	14.9	279.6	N	201.4	50.6	12.9	19.6
7	840	3	0.0	−39.9	3.0	1613.1	N	307.1	86.5	2.2	3.6
6	835	7	359.1	−41.3	4.2	198.7	N	289.7	87.4	3.1	5.1
5	780	6	349.4	−40.3	7.3	84.4	N	233.0	79.9	5.3	8.8
4	755	3	2.6	−43.6	14.2	76.0	N	22.9	87.4	11.0	17.7
3	710	3	342.4	−51.7	9.3	173.8	N	191.0	73.4	8.6	12.7
2	690	4	346.7	−21.1	7.1	167.5	N	265.1	70.2	3.9	7.5
1	655	4	331.6	−54.6	9.5	93.8	N	190.6	64.1	9.5	13.4

TABLE 15

Palaeomagnetic data for sequence PA(15)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
17	995	4	356.6	−56.5	7.8	136.7	N	140.4	78.5	8.2	11.3
16	930	12	165.7	64.8	5.0	75.0	R	331.7	−66.3	6.5	8.0
15	820	2	353.6	−20.5	16.4	231.2	N	284.0	73.4	9.0	17.2
14	790	3	332.6	−36.8	11.7	111.2	N	223.2	64.2	8.0	13.7
13	770	3	359.7	−39.1	5.7	454.4	N	302.5	86.1	4.1	6.8
12	720	6	358.1	−36.5	5.0	177.2	N	289.2	84.0	3.4	5.8
11	690	2	331.5	−49.9	28.8	76.8	N	199.3	64.5	25.7	38.4
10	685	4	352.4	−36.7	5.0	327.8	N	253.6	81.0	3.4	5.8
9	620	3	358.3	−46.7	3.3	1367.0	N	164.0	87.5	2.7	4.3
8	600	3	358.7	−25.6	10.9	127.4	N	300.8	77.4	6.3	11.8
7	590	3	358.8	−37.8	9.9	155.1	N	293.5	85.0	6.9	11.7
6	565	2	347.0	−27.1	13.2	356.3	N	257.8	73.1	7.8	14.4
5	500	4	8.8	−44.8	4.4	419.3	N	41.6	82.0	3.5	5.6
4	450	3	8.0	−44.8	8.9	189.3	N	45.0	82.7	7.1	11.2
3	425	4	3.4	−43.6	4.8	352.3	N	35.7	86.9	3.7	6.0
2	385	3	21.4	−17.0	6.0	412.0	N	0.6	63.5	3.2	6.2
1	360	4	6.0	−43.0	5.4	285.7	N	32.0	84.5	4.1	6.7

Nardy (1988) made a detailed investigation of these sequences and concluded on statistical bases that these flows are largely contemporaneous and that their mean magnetization direction differs significantly from the mean of the lower flows. These acid flows, however, may differ in age from those of the same type in the northern region, coded MI(18) here, which are of reversed polarity.

The mean direction of section PB(14), made up of all normal polarity flows, surprisingly differs from section PA(15) and is similar to the Chapecó normal mean. It is possible that this normal phase

TABLE 16

Palaeomagnetic data for sequence IC(16)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
12	805	5	158.3	34.3	10.9	49.8	R	49.9	−68.9	7.2	12.5
11	750	3	167.4	10.9	11.1	123.2	R	92.9	−66.7	5.7	11.3
10	710	3	342.2	−33.1	30.4	17.4	N	236.6	72.0	19.6	34.5
9	675	3	358.1	−30.7	8.9	189.0	N	294.9	81.0	5.5	9.9
8	668	3	353.1	−34.7	14.3	74.4	N	258.8	81.2	9.4	16.4
7	665	3	6.3	−29.2	24.9	25.3	N	339.1	78.6	15.2	27.5
6	630	5	354.8	−30.6	7.0	119.1	N	276.9	79.9	4.3	7.8
5	585	5	2.7	−44.7	10.8	50.6	N	59.2	87.3	8.6	13.6
4	530	3	359.4	−42.8	6.4	367.9	N	258.4	89.3	4.9	7.9
3	520	2	2.1	−46.0	3.9	3934.2	N	83.2	87.1	3.2	5.0
2	505	3	7.5	−62.6	10.3	144.1	N	110.0	70.3	12.6	16.1
1	500	3	7.0	−65.1	4.4	770.2	N	113.7	67.4	5.7	7.1

TABLE 17

Palaeomagnetic data for sequence PP(17)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
7	700	3	169.1	30.6	12.9	91.0	R	70.7	−78.0	8.0	14.4
6	580	2	168.7	47.9	38.6	43.8	R	6.8	−78.1	32.9	50.4
5	510	3	10.2	−63.4	3.9	989.3	N	111.7	66.5	4.9	6.2
4	480	3	7.5	−37.4	7.1	295.8	N	25.9	82.7	4.9	8.3
3	470	3	346.1	−35.0	8.4	213.5	N	232.6	76.6	5.6	9.7
2	460	5	354.4	−33.4	4.1	334.4	N	260.1	83.0	2.7	4.7
1	390	3	189.9	2.7	5.3	529.8	R	155.4	−66.5	2.7	5.3

TABLE 18

Palaeomagnetic data for sequence MI(18)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
4	535	3	176.7	21.5	9.5	169.2	R	115.6	−77.5	5.3	10.0
3	530	3	175.6	30.0	9.9	154.7	R	99.9	−81.7	6.1	11.0
2	515	3	172.2	30.1	18.4	45.6	R	83.1	−79.8	11.3	20.4
1	500	3	180.1	27.7	3.1	1552.6	R	131.3	−81.5	1.8	3.4

TABLE 19

Palaeomagnetic data for sequence MT(19)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
MT-8	520	3	18.0	−26.4	19.6	40.5	N	17.7	71.5	11.5	21.2
MT-7	420	2	187.2	29.5	26.7	89.4	R	180.4	−81.4	16.3	29.5
MT-6	340	3	177.9	28.3	11.7	111.4	R	104.8	−84.4	14.6	28.7
MT-5	510	3	359.6	−21.8	9.7	159.4	N	302.9	78.9	5.4	10.2
MT-4	355	3	347.3	−25.1	10.8	129.8	N	249.9	74.9	6.2	11.6
MT-3	330	2	181.9	43.6	38.6	43.9	R	276.9	86.3	30.0	48.1
MT-2	320	3	359.0	−34.9	10.5	138.5	N	288.4	86.8	7.0	12.1
MT-1	300	3	9.0	−29.6	6.5	359.6	N	3.1	79.6	4.0	7.2
MT-sill	600	3	7.9	−30.6	5.1	582.3	N	256.8	80.4	3.2	5.7

was present in section PA(15), since we still find a remanent normal flow in its upper part.

6. Secular variation

As pointed out previously, some of the southern sequences comprise several polarity intervals, which allow us to assume an elapsed time of the order of several secular variation cycles. In fact, analyses of the palaeosecular variation recorded in some Serra Geral lava successions (Pacca and Ernesto, 1982) have revealed a cyclicity in the magnetic directions on a time-scale of the order of 10^3–10^4 years, with data covering several cycles.

However, the angular dispersion that affects

TABLE 20

Palaeomagnetic data for sequence MG(20)

Flow number	Elevation (m)	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
		N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
MG-11	320	3	352.0	−33.6	12.0	105.8	N	233.2	82.3	7.8	13.7
MG-10	310	3	341.4	−40.8	11.4	117.0	N	207.2	72.5	8.4	13.8
MG-9	300	2	345.0	−33.3	57.1	21.2	N	226.6	75.7	36.9	64.9
MG-8	300	3	357.7	−35.7	4.0	942.5	N	254.8	87.3	2.7	4.6
MG-7	330	3	7.7	−36.8	12.7	95.1	N	42.2	82.6	8.7	14.9
MG-6	305	3	326.1	−44.0	10.1	147.1	N	202.4	58.5	7.9	12.6
MG-5	240	2	168.4	18.7	19.5	164.5	R	81.3	−74.3	10.6	20.3
MG-4	600	3	358.9	1.1	14.7	70.6	I	305.0	71.5	7.4	14.7
MG-3	560	3	169.7	18.2	12.0	105.4	R	82.7	−76.3	6.5	12.5
MG-2	475	2	129.7	5.6	21.3	138.7	R	50.4	−38.5	10.7	21.4
MG-1	420	3	5.6	−36.6	13.5	83.8	N	57.7	84.4	9.2	15.8

data from each sequence is frequently less than the expected dispersion due to secular variation (Ernesto, 1985). This is actually an artifact of the inhomogeneous distribution of data along the secular variation cycles, with implications for the accuracy with which pole positions are determined. This effect is probably more important in some of the short sequences from central and northern regions where, frequently, only the topmost flows are exposed.

7. Palaeomagnetic poles

Palaeomagnetic poles calculated in this paper are plotted in Fig. 8a (see p. 173). Creer's (1962) Serra Geral pole is not being considered in this paper, partly because many of the newer results include the lavas studied by him and partly because of the rigorous flow-wise sampling criterion adopted here. In Fig. 8a poles are discriminated according to the geochemical provinces (Piccirillo et al., 1988a) that they come from.

TABLE 21

Palaeomagnetic data for the intrusive rocks SD(21)

Intrusion number	Mean magnetization direction					Polarity	Virtual geomagnetic poles			
	N	D (degrees)	I (degrees)	α_{95} (degrees)	K		Longitude (°E)	Latitude (degrees)	d_p (degrees)	d_m (degrees)
S-1	2	140.8	35.7	11.8	444.9	R	39.0	−53.5	7.9	13.7
S-2	3	345.4	−35.4	8.3	218.7	N	230.4	76.1	5.5	9.6
S-3	2	351.5	−36.2	4.1	3618.1	N	239.0	81.7	2.8	4.8
S-4	5	346.2	−31.5	6.9	123.6	N	245.3	75.6	4.3	7.7
S-5	2	328.0	−43.8	31.3	65.7	N	210.2	60.8	24.4	39.1
S-6	3	163.5	21.9	10.2	146.5	R	75.4	−70.3	5.7	10.8
S-7	2	334.1	−45.7	10.0	623.5	N	206.6	66.2	8.1	12.7
D-1	3	154.4	52.4	7.5	265.5	R	13.4	−65.8	7.1	10.3
D-2	3	161.1	46.8	12.8	92.7	R	23.4	−72.5	10.6	16.5
D-3	3	350.7	−50.2	8.7	200.7	N	179.4	79.3	7.8	11.7
D-4	1	131.6	42.6	–	–	R	30.2	−46.1	–	–

TABLE 22

South palaeomagnetic poles for each data set. N is the number of flows or intrusions used for calculating the means; α_{95} and k are Fisher's (1953) statistical parameters

Section	N	Longitude (°E)	Latitude (°S)	α_{95} (degrees)	K
JS(1)	20	111.1	82.4	4.9	45
SM(2)	20	101.0	82.5	4.1	66
BV(3)	18	112.6	84.0	3.9	81
CV(4)	20	109.7	86.3	4.7	49
PH(5)	13	129.3	83.4	7.7	30
BM(6)	10	122.0	86.0	6.0	66
TA(7)	16	98.4	84.0	9.1	18
PC(8)	19	90.9	85.0	3.3	107
RA(9)	14	115.0	84.9	4.8	68
GB(10)	23	127.4	84.3	3.1	99
QI(11)	21	38.5	84.7	3.9	67
HI(12)	20	91.7	85.3	4.8	48
PU(13)	22	70.0	80.8	4.5	48
PB(14)	7	49.9	82.6	7.9	59
PA(15)	17	73.4	86.4	6.6	30
IC(16)	9	92.9	82.7	6.8	58
PP(17)	6	112.4	87.4	14.5	22
MI(18)	4	107.3	80.5	4.4	425
MT(19)	8	152.0	83.0	6.4	75
MG(20)	7	65.6	84.3	7.9	59
SD(21)	10	36.6	71.6	6.8	52

Poles from the southern Paraná Basin are well grouped and they approximate a circular distribution. However, poles from the other two regions are more scattered and impose an elongated pattern to the overall distribution. The SD pole, which corresponds to sills and dykes from the northeastern basin, is located significantly far from the main group, and its confidence circle overlaps only partially the PB(14) circle.

However, the palaeomagnetic poles become more tightly grouped (Fig. 8b) when poles QI(11), HI(12) and PU(13) are recalculated (Table 23) excluding VGPs related to the overlying acid Chapeco flows and their basalt intercalations. As mentioned in a previous section, those acid flows (and their associated basic flows) form a coherent group of VGPs; their mean corresponds to the CH pole in Fig. 8a, which is very close to pole PB(14). Therefore, the two normal polarity poles PB(14) and CH, seem to characterize a slightly younger phase of the Serra Geral volcanism, thus a pole path is delineated in Fig. 8b with decreasing ages towards the SD pole. We, therefore, will adopt three mean palaeomagnetic poles for the Serra Geral Formation, which are listed in Table 24 and plotted in Fig. 9 (see p. 173).

Pole SG1 corresponds to the average of 18 individual poles and is, therefore, a high-precision pole for the Lower Cretaceous of South America. In this calculation, poles PP(17) and MI(18), which are the least accurate, being based on only six and four observations respectively, have been combined in a unique pole in order to give lower weight to them. Poles QI(11), HI(12) and PU(13) were taken from Table 23. Pole SG2 is the average between the formerly named PB and CH poles and SG3 coincides with the SD pole.

The ages assigned to each of these poles are in the range 140–120 Ma given by the available radiometric ages (see references above). Piccirillo et al. (1987) report K–Ar ages on feldspar concentrates from samples collected near the locality of Piraju, northern Paraná Basin, which indicate that the Chapecó acid volcanics and the overlying basalts were erupted around 135 Ma. As these samples come from the same locality as our samples MI(18), we therefore consider 135 Ma as a reliable age for pole SG1. Mantovani et al. (1985) obtained Rb–Sr ages for three Chapeco-type samples from central Paraná Basin, which range from 131.8 ± 5.9 to 138.4 ± 3.5 Ma. Although the authors claim that the best age estimate is 135.4 ± 3.5 Ma, we feel that the lower limit, i.e. 132 Ma is more convenient to account for pole differences.

Radiometric dates that can be related to pole SG3 are few. For the Ponta Grossa dykes, four determinations give ages ranging from 144 to 114 Ma (Amaral et al., 1966; J.P. Pinese, personal

TABLE 23

Recalculated palaeomagnetic poles for some sections in the central Paraná Basin when the rhyodacite Chapecó flows (CH) are excluded

Section	N	Longitude (°E)	Latitude (°S)	α_{95} (degrees)	K
QI(11)	9	137.9	87.5	5.1	102
HI(12)	15	113.4	85.2	5.4	51
PU(13)	10	94.5	82.3	7.1	47
CH	21	27.4	81.0	3.9	66

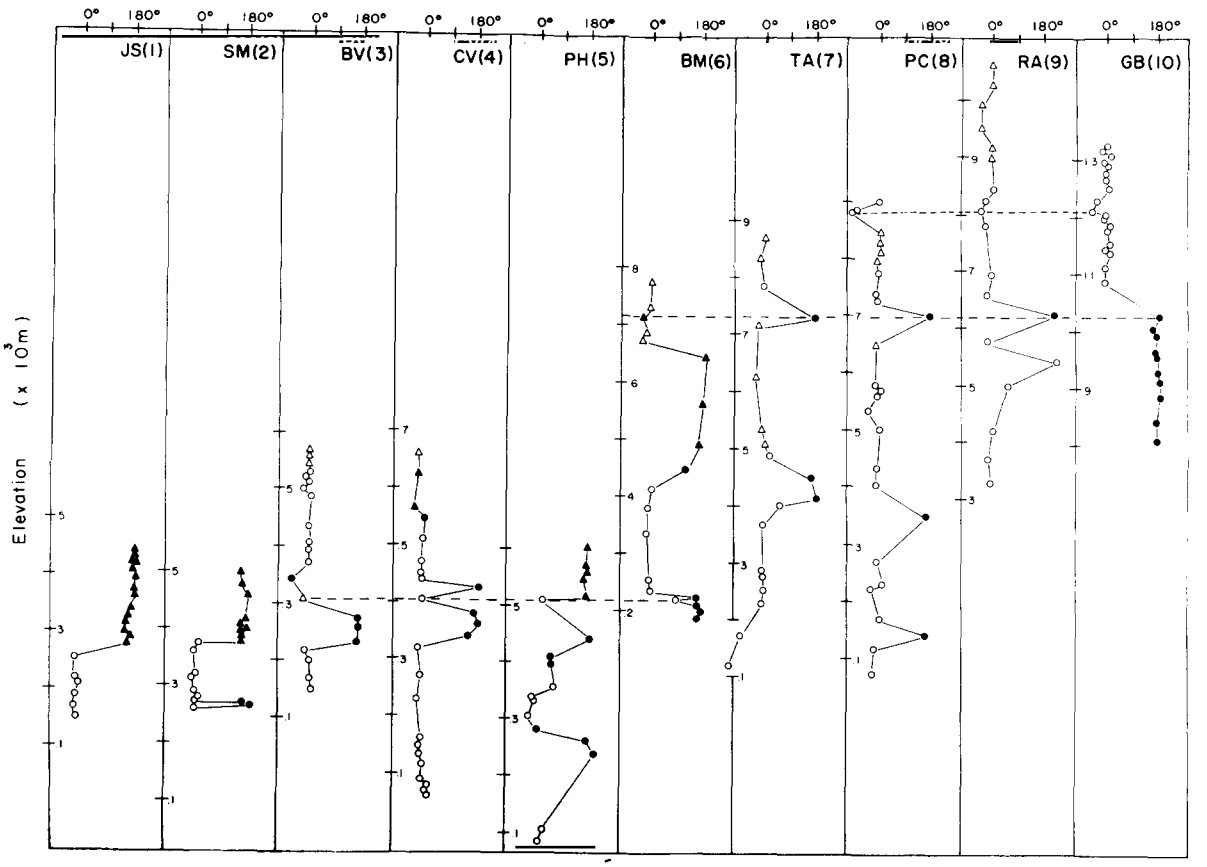


Fig. 4. Magnetic declination variations in the sections from southern Paraná Basin. Solid symbols denote positive inclinations, open symbols negative inclinations. Circles correspond to basic lavas and triangles are Palmas acid lavas. Sequences have been adjusted in order to fit the common features they display (broken lines).

communication). Amaral et al. (1966) also report a strong peak close to 120 Ma in a histogram of ages for 15 dykes and sills from the Paraná Basin, out of which 12 come from the northeastern portion of the basin. This figure can be taken as an upper limit for SG3 age since this pole must be older than the lower boundary of the 'Cretaceous normal magnetic interval' (~ 118 Ma according to Kent and Gradstein, 1985). Pole SG3 was computed from both normal and reversed polarity directions.

8. Lower Cretaceous polar shift

The mean palaeomagnetic poles calculated for the Serra Geral Formation (Fig. 9) define a shifting path corresponding to the time interval 135–120 Ma, approximately. This path differs considerably from those proposed, for example, by Valencio et al. (1983) and Irving and Irving (1982), however, it is supported by other South America poles for Upper Jurassic to Lower Cretaceous.

Selected poles (Table 25) ranging from ~ 165 to 120 Ma are also plotted in Fig. 9 and coded according to Valencio et al. (1983). Poles from sites that possibly underwent tectonic movements were discarded. Poles J1, J2 and K3 (triangles) are older than 130 Ma and tend to be close to the Serra Geral pole SG1, excepting for K3. This pole corresponds to the Rio Los Molinos dykes, which have radiometric ages ranging from 150 ± 10 to 129 ± 10 Ma and also from 68 ± 5 to 63 ± 5 Ma (Linares and Valencio, 1975). These younger ages

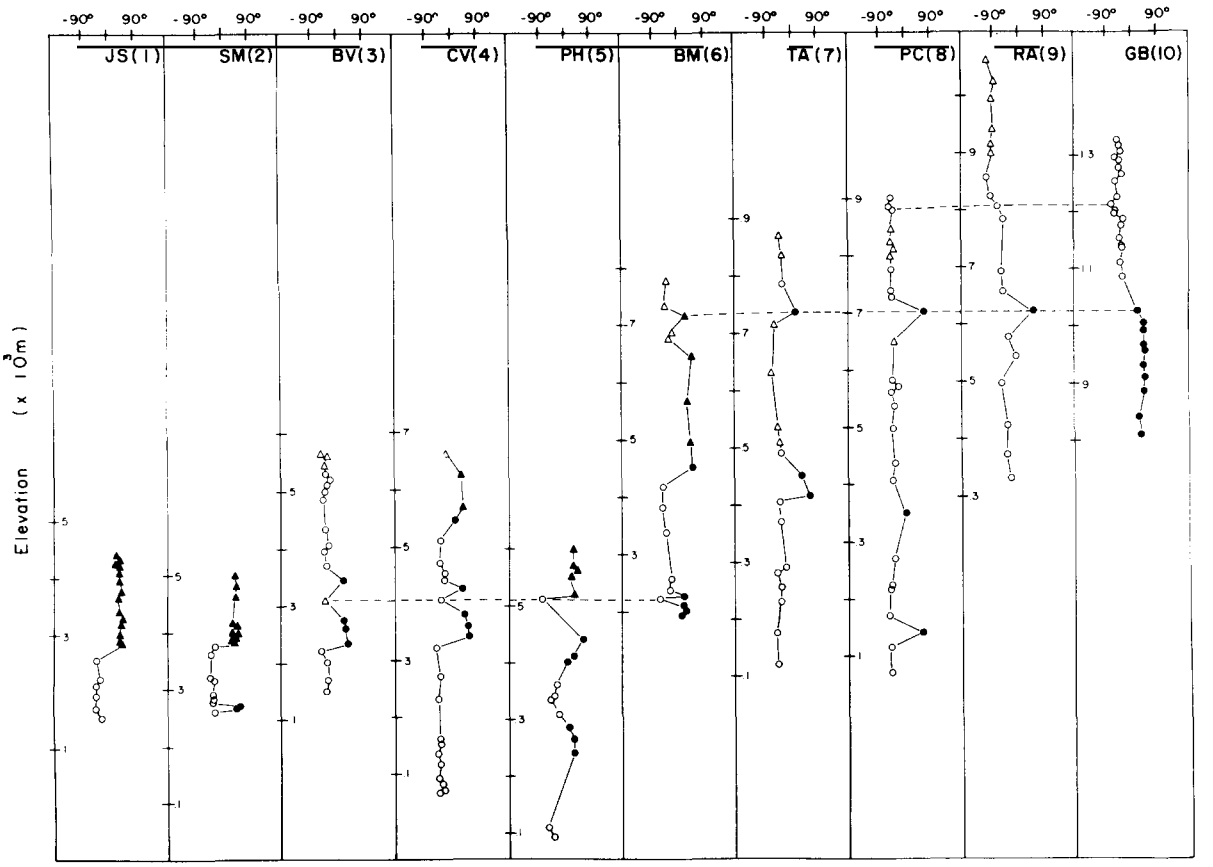


Fig. 5. Magnetic inclination variations in the sections from the southern Paraná Basin. Explanation as in Fig. 4.

have been attributed to argon loss. However, considering the large scatter in radiometric data and the uncertainty (10 Ma) in the older ages, it is reasonable to admit that these dykes may be younger than 130 Ma. Poles younger than 130 Ma (squares), if K13 is not considered, tend to lie along the Serra Geral pole path and closer to SG2 and SG3 poles. The pole K13, to which a mean age of 118 Ma was attributed by Schult and Guerreiro (1979), is very close to the older pole SG1, although their circles of confidence do not overlap. In fact, radiometric ages as old as 199 Ma (range: 199–110 Ma) are reported by Caldasso and Hama (1978) for the diabase intrusions crop-

TABLE 24

Mean palaeomagnetic poles for the Serra Geral Formation

Code	Mean age (Ma)	$N/(n)$ [a]	Latitude (°S)	Longitude (°E)	α_{95}	K	Poles included in the means
SG1	~135	13	85	108	1.1	1038	1/13, 15/20
SG2	~131	2	82	38	7.8	1039	14, CH
SG3	<118	(10)	72	37	6.8	52	SD

[a] N is the number of sequences included in the means; n is the number of intrusions included in the mean.

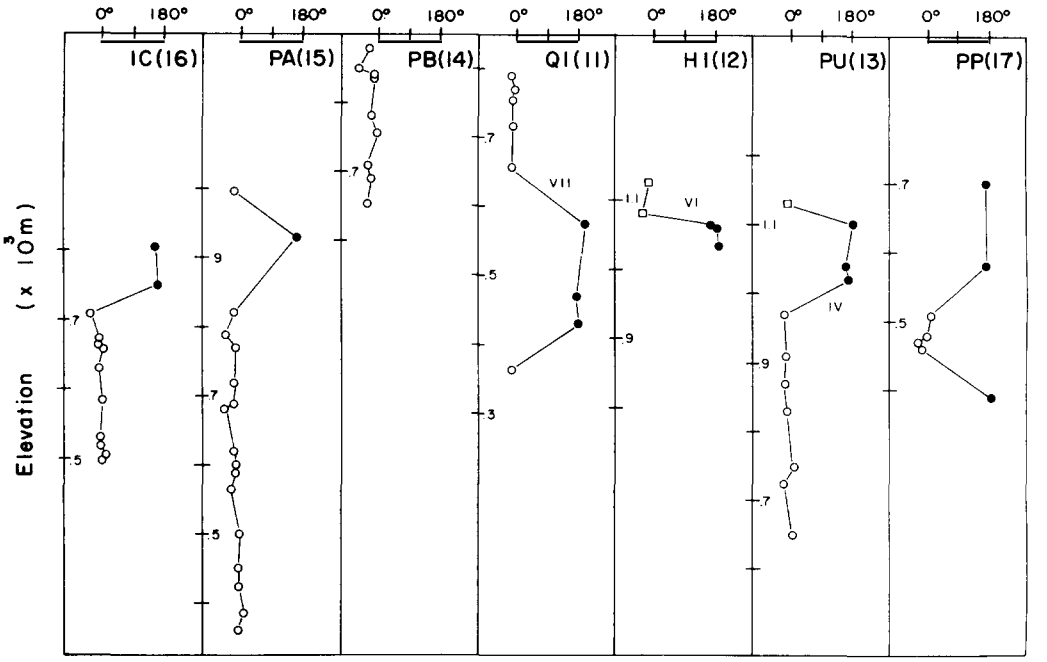


Fig. 6. Magnetic declination variations in the sections from the central and northern Paraná Basin. Only the thickest QI(11), HI(12) and PU(13) blocks are represented. Squares denote the Chapecó acid lavas. Other explanation as in Fig. 4.

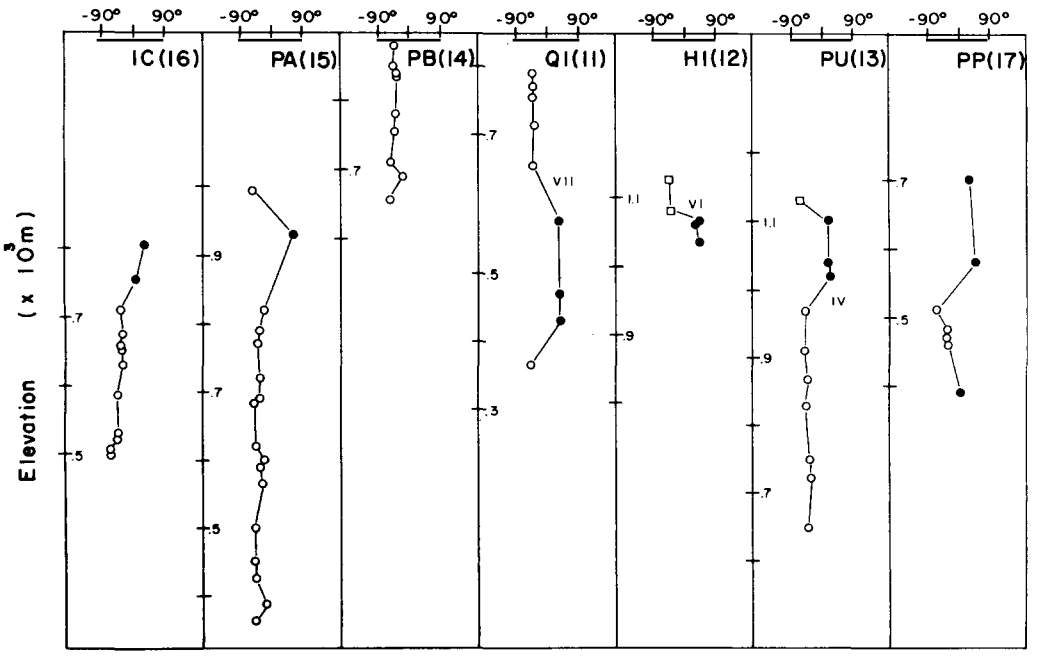


Fig. 7. Magnetic inclination variations in the sections from the central and northern Paraná Basin. Explanation as in Figs. 4 and 6.

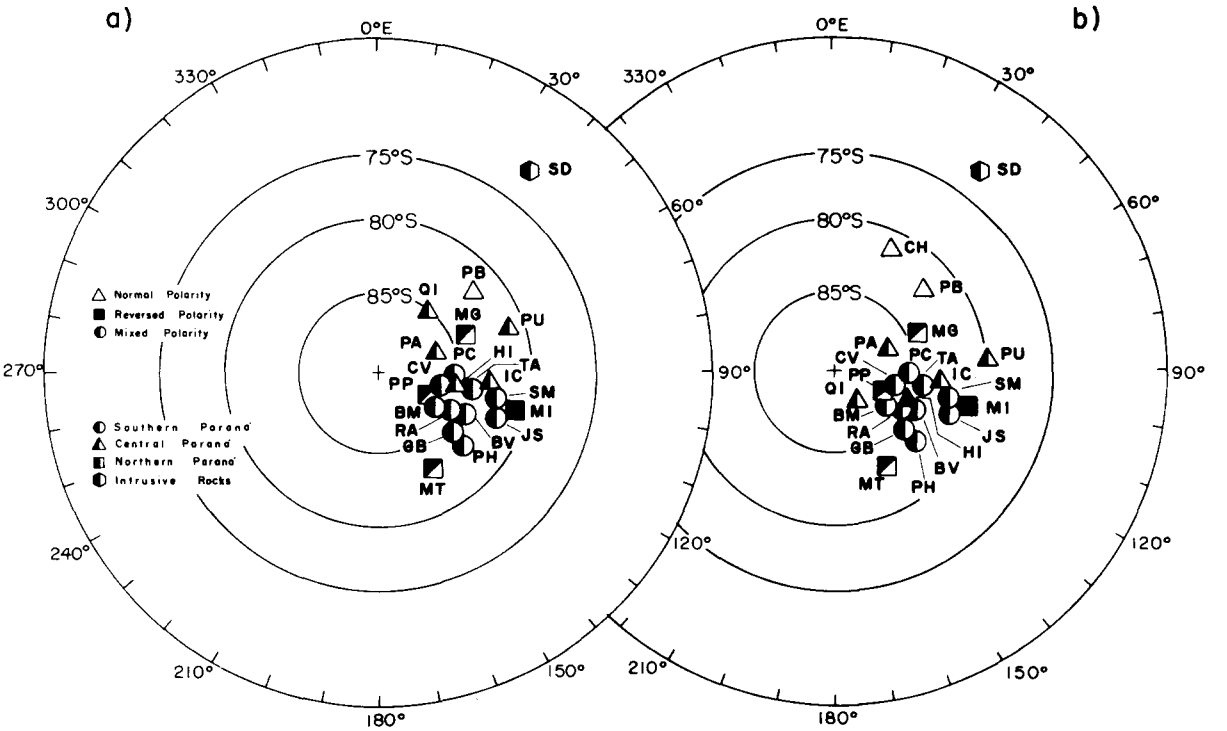


Fig. 8. (a) Palaeomagnetic poles for each volcanic sequence and for intrusions. (b) Same as in (a) but with QI(11), HI(12) and PU(13) poles recalculated in order to exclude the Chapecó acid flows (CH).

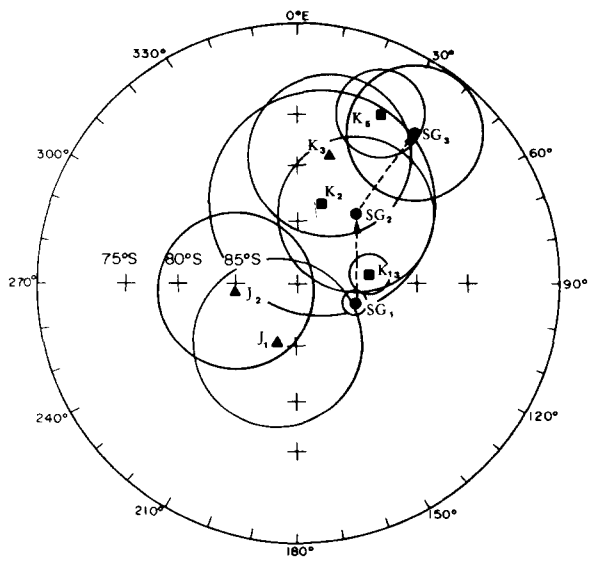


Fig. 9. Mean palaeomagnetic poles for the Serra Geral Formation (circles) and selected Mesozoic poles for South America. Triangles indicate poles of age >130 Ma and squares refer to ages <130 Ma. Confidence circles (α_{95}) are also plotted.

ping out in the eastern border of the Parnaiba (Maranhao) Basin (pole SAK13). These authors have listed 18 ages, out of which 12 are > 125 Ma and only six are lower or equal to 125 Ma. It is therefore clear that an older age, possibly comparable to the age of SG1, should be assigned to pole SAK13.

The Lower Cretaceous path based on the Serra Geral mean poles proposed here indicates that during the time interval between poles SG1 and SG2, the South America platform moved southwards ~ 5° and by the time of pole SG3 it had completed a clockwise rotation of ~ 10°. Since the Serra Geral volcanism is intimately associated with the South Atlantic opening (e.g. Piccirillo et al., 1988b), this clockwise rotation is consistent with the predicted rotations for a south to north Atlantic opening model (e.g. Martin et al., 1981; Vink, 1982).

TABLE 25

Selected Jurassic and Cretaceous palaeomagnetic poles for South America

Formation	Age (Ma)	Site		Palaeomagnetic pole			Code	References
		Latitude (°S)	Longitude (°E)	Latitude (°S)	Longitude (°E)	α_{95}		
Chon Aike	166 ± 5	44–48	65–69	85	197	6	SAJ1	Vilas (1974)
Maranhão Volcanics	158 ± 12	6.4	47.4	85	263	7	SAJ2	Schult and Guerreiro (1979)
Rio Los Molinos dykes	150–129							
	68– 63	31.6	64.5	78	13	8	SAK3	Linares and Valencio (1975)
Almafuerte lava flows	130–120	32.2	64.3	72	25	6	SAK5	Mendia (1978)
Cerro Colorado	121	32.2	64.1	83	16	10	SAK2	Valencio (1972); Valencio and Vilas (1976)
Maranhão intrusions	118	6.5	42.0	84	81	1.9	SAK13	Schult and Guerreiro (1979)

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